MILITARY SPECIFICATION

VISORS, SHATTER RESISTANT

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for polycarbonate visors for the EEK-4A/P Helmet Visor Assemblage.

1.2 Classification. Visors shall be of the following classes, see 6.2:

Class 1 Clear Visor
Class 2 Neutral Visor

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-393	-Plastic Molding Material, Polycarbonate, Injection and Extrusion
PPP-B-636	-Box, Shipping Fiberboard
PPP-B-676	-Boxes, Setup

MILITARY

MIL-V-22272	-Visor, Protective, Helmet

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, ESSD, Code 93 Lakehurst, New Jersey 08733 by using the self-addressed Standard Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-406	-Plastic, Methods of Testing

MILITARY

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-Marking for Shipment and Storage
MIL-STD-130	-Identification Marking of U.S. Military Property
MIL-STD-810	-Environmental Test Methods

DRAWINGS

NAVAL AIR DEVELOPMENT CENTER

954AS410	-Visor Lens Subassembly, Medium
954AS411	-Visor Lens Subassembly, Large

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

Z58.7.1-1951	-Method of Spectrophotometric Measurement for for Color

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 412	-Tension Testing of Vulcanized Rubber

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified in the contract or order, the visors furnished in accordance with this specification shall be a product which has been inspected and has passed the first article inspection specified herein, (see 4.3).

3.2 Materials. The visors and guides shall be molded from a polycarbonate resin as specified in L-P-393.

3.3 Design and construction. The design and construction of the visors shall be as specified in NADC Drawing 954AS410 or 954AS411 as applicable.

3.3.1 Areas of vision. The visor area of vision shall be classified as critical and non-critical. The critical area of vision shall be as specified on the applicable drawing and shall conform to the requirements and optical characteristics specified herein. The non-critical area of vision shall be all other areas of the visor not indicated as critical and shall conform to the same requirements as for the critical area insofar as the state-of-art permits.

3.3.2 Point "C". Point "C" shall be located as specified in the applicable NADC drawing 954AS410 or 954AS411.

3.4 Performance.

3.4.1 Visible transmittance.

3.4.1.1 Class 1-Clear visor. The total visible transmittance of Class 1 visors, when calculated as specified, (see 4.6.2) shall be substantially uniform over the entire visor and shall be not less than 87 percent at point "C" as defined on applicable drawings.

3.4.1.2 Class 2-Neutral visor. The total visible transmittance of Class 2 visors at point "C", when calculated as specified, (see 4.6.2) shall be 12±4 percent. The transmittance at the top of the visor shall not exceed the transmittance at the bottom of the visor. The transmittance at the left point "C" and right point "C" shall not differ by more than 20 percent of the higher of the two values.

3.4.2 Neutrality.

3.4.2.1 Class 2 visor. The visible transmittance of the Class 2 visor, when calculated as specified, (see 4.6.3) shall vary with wavelength between 430 and 730 nanometers so that the average percentage deviation from neutrality is less than 12 percent.

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3.4.3 Chromaticity.

3.4.3.1 Class 2 visor. The chromaticity coordinates, x and y, of the Class 2 visor, when calculated as specified, (see 4.6.4.1) shall plot within the ellipse in Figure 1.

3.4.4 Diffuse transmittance (all classes of visors). The diffuse transmittance of the visor, when determined as specified, (see 4.6.5) shall not exceed 5 percent of the total visible transmittance.

3.4.5 Haze. The haze value of the visor shall not exceed 2.0 percent when tested as specified, (see 4.6.6).

3.4.6 Ultraviolet transmittance.

3.4.6.1 The average ultraviolet transmittance of the Classes 1 and 2 visors, when determined as specified, (see 4.6.7) shall be less than 5 percent in the erythemal band (290-320 nanometers).

3.4.7 Abrasion resistance. After being subjected to the abrasion tests specified, (see 4.6.8) the increase in haze shall not exceed 6 percent and the decrease in transmittance shall not exceed 4 percent.

3.4.8 Adhesion of coating. The coating, when tested as specified, (see 4.6.9) shall not be removed, dislodged, or affected in any way. Removal or loosening of the coating shall be cause for rejection.

3.4.9 Prismatic deviation (all classes of visors).

3.4.9.1 Vertical prismatic deviation. The vertical prismatic deviation between point "C" for the right eye and point "C" for the left eye, when measured as specified, (see 4.6.10.1) shall not differ by more than 0.25 diopters. In addition, there shall be no abrupt transitions in prismatic deviation throughout the critical and semi-critical area of the visor.

3.4.9.2 Horizontal prismatic deviation. The sum of the horizontal prismatic deviation at point "C" for the right eye and point "C" for the left eye, when measured as specified, (see 4.6.10.1) shall not exceed 0.75 diopters. The difference between the horizontal prismatic deviation at point "C" for the right eye and point "C" for the left eye shall not exceed 0.18 diopters. In addition, there shall be no abrupt transitions in prismatic deviation throughout the critical and semi-critical area of the visor.

3.4.10 Spherical and cylindrical power (all classes of visors). The spherical and cylindrical power, when measured, as specified, (see 4.6.11) shall not exceed plus or minus 0.125 diopters at point "C".

3.4.11 Critical vision area distortion (all classes of visors). The distortion within vision area, when examined as specified, (see 4.6.12) shall be within the acceptable limits shown in Figure 2.

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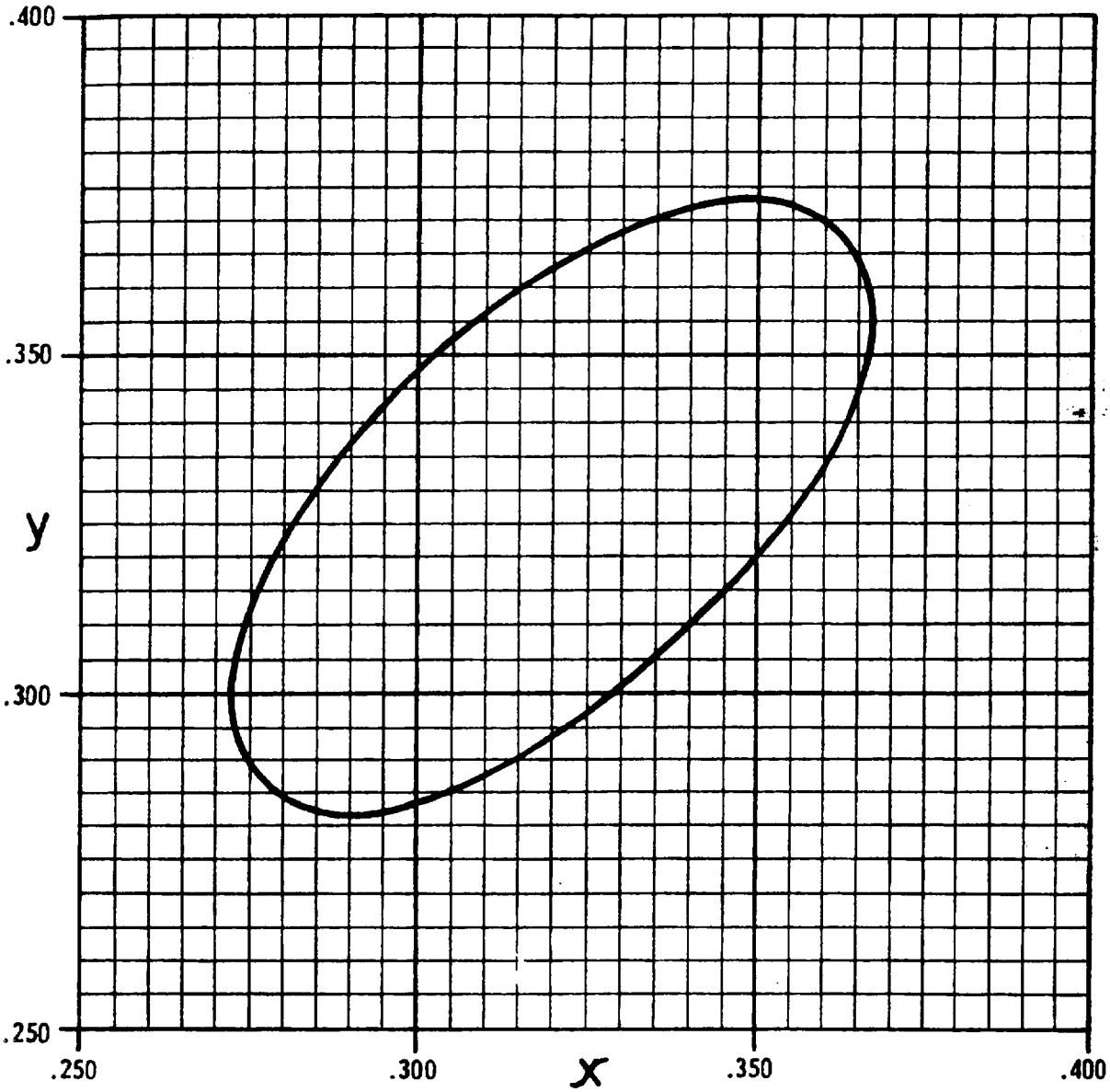


FIGURE 1. Munsell value 4, chroma 2 ellipse.

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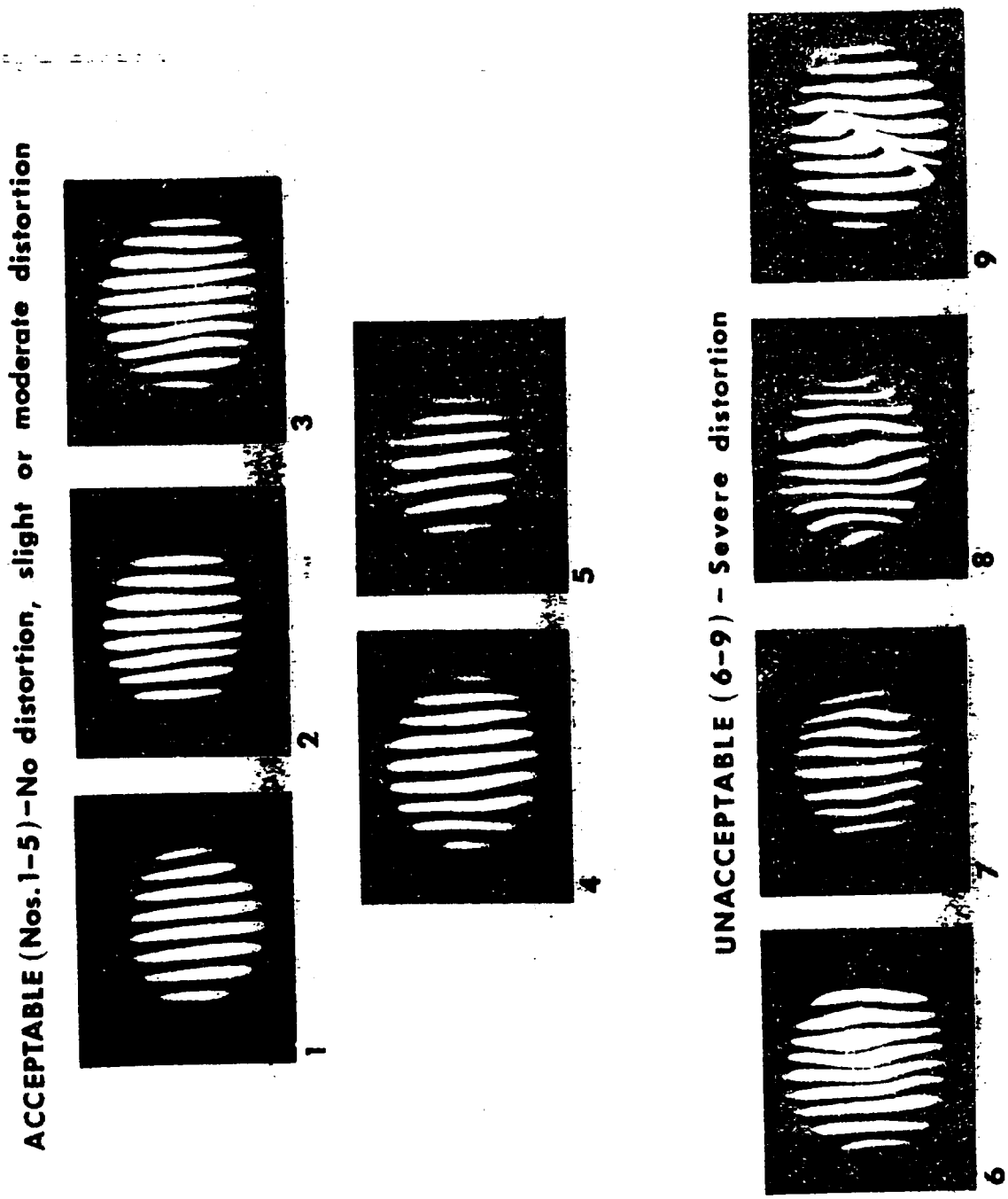


FIGURE 2. Distortion standards.

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3.4.12 Crack propagation resistance (thermal relocation method). When tested as specified, (see 4.6.13) the resistance to crack propagation shall be determined in terms of percentage of recoverable stretch in the visor and shall be a minimum of 30 percent.

3.4.13 Impact resistance. The visor, when tested as specified, (see 4.6.14) shall not fracture. A visor shall be considered to have factured if it cracks (through its entire thickness) into two or more separate pieces or if any material visible to the naked eye is detached from the ocular surface.

3.4.14 Penetration resistance. The visor shall not be pierced when subjected to the penetration test specified (see 4.6.15).

3.4.15 Visual examination for other optical defects (all classes of visors). The critical and semi-critical area of the visor shall be free from visually detectable (with the unaided eye) striae, waviness, cloudiness and imperfections such as pits, lint, bubles, scratches, and foreign particle imperfections which are gross in size and quantity.

3.5 Workmanship. All visors shall be constructed to produce an item free from all defects which would affect proper functioning in service. The visors shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not render the visors unacceptable for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such, inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the visor are classified as follows:

a. First article inspection. First article inspection consists of all examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this spectification. (See 4.3.)

b. Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the

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requirements set forth in this specification. The Government reserves the right to require the manufacturer to submit an occasional sample from a production lot for inspection by the government to assure that the product continues to meet the requirements stated herein. (See 4.4)

4.3 First article inspection. The first article inspection shall consist of all examinations and tests listed herein. (See 3.1)

4.3.1 First article samples. Unless otherwise specified, as soon as practicable after award of the contract or order, the manufacturer shall submit eight first article samples of each class. The samples shall be representative of the construction, workmanship and materials to be used during production. When a manufacturer is in continuous production of these visors from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the procuring activity (See 6.2). Approval of the first article samples or the waiving of first article inspection does not preclude the requirement of submitting to the quality conformance inspection. Two first article samples of each class shall be forwarded to the Supply Officer, Naval Air Development Center, Crew Systems Department, Warminster, PA 18974, Code 4041, and six samples of each class shall be forwarded to the Naval Submarine Medical Research Laboratory, Naval Submarine Medical Center, Naval Submarine Base New London, Groton, Connecticut 06340, Attention: Vision Branch. The samples shall be plainly marked with the following information:

Samples submitted by (name) (date) for first article inspection in accordance with the requirements of this specification under Contract No.

4.3.1.1 Upon the completion of the first article inspection, all the applicable inspection reports, and when applicable, recommendations and comments pertinent for use in monitoring production shall be forwarded to the Government quality control representative. One approved first article sample shall be returned to the manufacturer for use in monitoring production. The remaining samples shall be consumed or destroyed in the first inspection and shall not be considered as a part of the quantity to be delivered under contract.

4.4 Quality conformance inspection. The sampling and inspection levels shall be as specified in MIL-STD-105. Quality conformance inspection shall consist of the following examinations and tests:

Visual examination (see 4.6.1)
Visible transmittance (see 4.6.2)
Critical vision area distortion (see 4.6.12)
Packaging (see 5.1)

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4.4.1 Sampling.

4.4.1.1 Inspection lot. An inspection lot shall be expressed in units of one class visor made essentially under the same conditions and from the same materials. The sample unit shall be one class visor of one size.

4.4.1.1.1 Packaging. An inspection lot size shall be expressed in units of one fully prepared shipping container, containing one class visor, fully prepared for delivery. The sample unit shall be one shipping container, containing one class visor, fully prepared for delivery with the exception it need not be sealed.

4.4.1.2 Sampling for tests and examinations of the visors. The sample size, acceptance criteria, examinations, and tests required for the visor shall be as specified in TAble I.

4.5 Test conditions.

4.5.1 Cleanliness. Before and after subjecting the visor to any test, the visor shall be thorcughly and carefully cleaned to remove dirt, oil, fingermarks, etc. using a mild detergent and water and dried with a soft cloth or visor tissue in order to assure that the following tests are performed on a clean visor.

4.6 Inspection methods.

4.6.1 Visual examination. Every visor shall be examined visually for critical and major defects to determine conformance with this specification. Each visor selected as a sample unit from the lot shall be thoroughly checked for dimensional conformance and visually examined for minor defects to determine conformance to this specification. The classification and list of defects, (see tables (II and III) shall be used to classify and enumerate the defects found.

4.6.1.1 Dimensions. Every visor shall be checked dimensionally to determine conformance to the dimensions specified herein and on the applicable drawing.

4.6.1.2 Packaging. Each of the fully prepared shipping containers, containing visors of one type and class, selected as a sample unit from the lot shall be visually examined to determine that the packaging, packing, and marking conform to this specification. The list of defects, (see Table III), shall be used to enumerate the defects found.

4.6.1.3 Marking. Markings shall be as specified in MIL-STD-130. Ink, color, size and location shall be as indicated on the applicable drawing. Letters and numerals of 1/4-inch height or less may not be stenciled. MIL-V-22272 is applicable for Class 2 visors.

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TABLE 1. Sample size, acceptance criteria, examinations and tests.

Inspection	Type of inspection	Paragraph		Sample size	Acceptance criteria
		Reference	Test		
Visual examination	Critical and minor	3.3.1 and 3.4.17	4.6.1	Every visor for critical defects and Inspection Level II for minor defects	Reject all units with any criteria defects and an acceptable quality level of 6.5 defects per 100 units for minor defects
Visible transmittance	Critical	3.4.1	4.6.2	1/ 2/	Acceptance number zero, rejection number one
Neutrality	Major	3.4.2	4.6.3	1/ 2/	Acceptance number zero, rejection number one
Chromaticity	Major	3.4.3	4.6.4	1/ 2/	Acceptance number zero, rejection number one
Diffuse transmittance	Major	3.4.4	4.6.5	1/ 2/	Acceptance number zero, rejection number one
Abrasion resistance (Scratch-resistant coatings on Class 1 and 2)	Critical	3.4.7	4.6.8	Inspection Level I 3/	Acceptance number zero, rejection number one
Adhesion of coating (Scratch-resistant coatings on Class 1 and 2)	Critical	3.4.8	4.6.9	Inspection Level I 3/	Acceptance number zero, rejection number one
Prismatic deviation	Critical	3.4.9	4.6.10	1/ 2/	Acceptance number zero, rejection number one
Spherical and cylindrical power	Critical	3.4.10	4.6.11	1/ 2/	Acceptance number zero, rejection number one
Critical vision area distortion	Critical	3.4.11	4.6.12	1/ 2/	Acceptance number zero, rejection number one
Visual examination for other optical defects	Major	3.4.17	4.6.1	1/ 2/	Acceptance number zero, rejection number one
Crack propagation resistance of visors	Critical	3.4.12	4.6.13	1/ 2/	Acceptance number zero, rejection number one

1/ The sample size shall be based on the applicable sample size code letter corresponding to the specified inspection level of MIL-STD-105.

2/ The sample size for all types of Classes 1, and 2 visors shall be determined as specified in MIL-STD-105, Inspection Level II.

3/ This is a destructive test (see 6.2).

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TABLE II. Classification of defects for visual examination of the visor.

Critical	Minor
1. Material imperfections - foreign matter embedded, striae, waviness, cloudiness, pits, bubbles or scratches. 2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws. 3. Any component missing, malformed, fractured, or otherwise damaged. 4. Any component loose or otherwise not securely retained. 5. Incorrect assembling or improper positioning of components. 6. Faulty workmanship or other irregularities. 7. Dimensions - any dimensions not as specified herein and on the applicable drawing.	201. Marking - missing, insufficient, incorrect, illegible, or not permanent. 202. Sharp edges on any part.

TABLE III. List of defects for packaging.

Examine	Defect
Contents or weight	Contains visors of more than one class; gross or net weight exceeds the requirements.
Exterior and interior markings	Missing, incorrect, incomplete, illegible; or improper size, location, sequence, or method of application; markings not the same on the interior and exterior container.
Material	Not as specified.
Workmanship	Inadequate application of the components such as incomplete closure of the container flaps, barrier material, bulging or distortion of the containers.

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4.6.2 Visible transmittance. Total visible transmittance is calculated from the spectrophotometric curve. Method for calculation is specified in Table IV. The spectrophotometric curve shall be obtained as specified in the ANSI Method of Spectrophotometric Measurement for Color Z58.7. When using the Hardy spectrophotometer, the sample shall be placed with the convex side toward the integrating sphere. Total visible transmittance readings may be obtained on Class 1 and 2 visors by using a Weston Foot Candle Meter No. 614, or equal, in conjunction with a Roundel Daylight Filter (Illuminant C) Type ADG-10, or equal, and a voltage stabilizer. If the total visible transmittance is a borderline pass or a borderline fail, the spectrophotometric method shall be used for final determination of acceptability of product. The visor shall pass the requirements specified in 3.4.1.

TABLE IV. Computation table for obtaining visible transmittance and chromaticity coordinates, with a neutral visor used as an example.

Wavelength (Nanometers)	T	XE	X (T·xE)	uE	Y (T·yE)	zE	Z (T·zE)
380	0.035	4	.140			20	.700
390	0.040	19	.760			89	3.560
400	0.047	85	3.995	2	.094	404	18.988
410	0.052	329	17.108	9	.468	1570	81.640
420	0.058	1258	71.804	37	2.146	5949	345.042
430	0.063	2997	188.811	122	7.686	14628	921.564
440	0.069	3975	274.275	262	18.078	19938	1375.722
450	0.075	3915	293.625	443	33.225	20638	1547.850
460	0.080	3362	268.960	694	55.520	19299	1532.920
470	0.085	2272	193.120	1058	89.930	14972	1272.620
480	0.091	1112	101.192	1618	147.238	9461	860.951
490	0.096	363	34.848	2358	226.368	5274	506.304
500	0.101	52	5.252	3401	343.501	2864	289.264
510	0.106	89	9.434	4833	512.298	1520	161.120
520	0.110	576	63.360	6462	710.820	712	78.320
530	0.113	1523	172.099	7934	896.542	388	43.844
540	0.116	2785	323.060	9149	1061.284	195	22.620
550	0.116	4282	496.712	9832	1140.512	86	9.976
560	0.114	5880	670.320	9841	1121.874	39	4.446
570	0.112	7322	820.064	9147	1024.464	20	2.240
580	0.109	8417	917.453	7992	871.128	16	1.744
590	0.108	8984	970.272	6627	715.716	10	1.080
600	0.107	8949	957.543	5316	568.812	7	.749
610	0.106	8325	882.450	4176	442.656	2	.212
620	0.107	7070	756.490	3153	337.371	2	.214
630	0.110	5309	583.990	2190	240.900		
640	0.114	3693	421.002	1443	164.502		
650	0.120	2349	281.880	886	106.320		
660	0.125	1361	170.125	504	63.000		
670	0.129	708	91.332	259	33.411		

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TABLE IV. Computation table for obtaining visible transmittance and chromaticity coordinates, with a neutral visor used as an example (Continued).

Wavelength (Nanometers)	T	XE	X (T·xE)	uE	Y (T·yE)	zE	Z (T·zE)
680	0.129	*369*	47.601	*134*	17.286		
690	0.127	*171*	21.717	*62*	7.874		
700	0.125	*82*	10.250	*29*	3.625		
710	0.128	*39*	4.992	*14*	1.792		
720	0.139	*19*	2.641	*6*	.834		
730	0.156	*8*	1.248	*3*	.468		
740	0.176	*4*	.704	*2*	.352		
750	0.194	*2*	.388	*1*	.194		
760	0.209	*1*	.209	*1*	.209		
770	0.210	*1*	.210				
Sums:			10131.436		10968.498		9094.690

Sum of XYZ = 30194.624

NOTES:

(a) The computation table is in *script*; the example data are in light print.
(b) T = Transmittance readings obtained on sample visor on spectrophotometer at 10 nonometer intervals.
(c) Tc = Total visible transmittance = $\Sigma Y/100{,}000$; sample visor = 10968.498/100,000 = 0.1097 = 10.97 percent.
(d) x (chromaticity coordinate) = $\Sigma X/\Sigma XYZ$; sample visor = 10131.436/30194.624 = .336
(e) y (chromaticity coordinate) = $\Sigma Y/\Sigma XYZ$; sample visor = 10968.498/30194.624 = .363

4.6.3 Neutrality.

4.6.3.1 Class 2 visor. The neutrality shall be calculated by the Judd Daylight Duplication Method (Shortened Form). Method of calculation is specified in Table V. The visor shall pass the requirements specified in 3.4.2.

4.6.4 Chromaticity.

4.6.4.1 Class 2 visor. The chromaticity coordinates shall be calculated from the spectrophotometric curve. Method of calculation is specified in Table IV.

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TABLE V. Worksheet for calculation of the average percent deviation from neutrality, with a neutral visor used as an example.

Wavelength (Nanometers)	T	Band n	Wavelength Range	Average, Transmittance Tn	Percent Deviation 100 (1-Tn/Tc) x Wt = Prod.		
430	0.063						
440	0.069						
450	0.075						
460	0.080	1	430-490	0.0799	27.2	5	136.0
470	0.085						
480	0.091						
490	0.096	2	460-520	0.0957	12.8	10	128.0
500	0.101						
510	0.106						
520	0.110	3	490-550	0.1087	.9	10	9.0
530	0.113						
540	0.116						
550	0.116	4	520-580	0.1134	3.4	10	34.0
560	0.114						
570	0.112						
580	0.109	5	550-610	0.1102	.5	10	5.0
590	0.108						
600	0.107						
610	0.106	6	580-640	0.1082	1.4	10	14.0
620	0.107						
630	0.110						
640	0.114	7	610-670	0.1156	5.4	10	54.0
650	0.120						
660	0.125						
670	0.129	8	640-700	0.1249	13.9	5	69.5
680	0.129						
690	0.127						
700	0.125	9	670-730	0.1318	20.1	1	20.1
710	0.128						
720	0.139						
730	0.156						
Sum						71	469.6

NOTES:

(a) The worksheet data are in *script*; the example data are in light print.

(b) T = Transmittance readings obtained on sample lens on spectrophotometer at 10 nanometer intervals.

(c) Tn = Average transmittance of 60 nanometer band. The Tn for a given band is the average of the seven tabulated values within that band, which is computed by dividing by 2 the first and last tabulated values in the band and adding to these two figures the five remaining tabulated values in the band and dividing this total sum by 6.

(d) Tc = Total visible transmittance of sample, visor, which is 0.1097 (see Table V).

(e) Average Percent Deviation from Neutrality = Sum of the Products/Sum of the Weights; sample visor = 469.6/71 = 6.61.

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4.6.5 Diffuse transmittance (all classes of visors). The diffuse transmittance shall be determined as specified in Method 3031 of FED-STD-406. Diffuse transmittance may also be measured on the Hardy Spectrophotmeter by taking a curve with the sample placed against the window of the integrating sphere and another curve with the sample placed at the entrance to the transmission compartment; the difference between these two curves defines the percent diffuse transmittance of the sample. The Cary Spectrophotometer with the Diffuse Reflectance Accessory may also be used. The visor shall pass the requirements specified in 3.4.4.

4.6.6 Haze. The haze shall be determined in accordance with method 3022 of Federal Standard No. 406. The visor shall pass the requirements specified. (See 3.4.5)

4.6.7 Ultraviolet transmittance. The ultraviolet transmittance shall be determined with an appropriate spectrophotometer (Cary, Beckman, etc.). Transmittance readings shall be obtained at 290, 300, 310, and 320 nonometers and these readings shall be summed and then divided by 4 to obtain the average ultraviolet transmittance of the sample. The visor shall pass the requirements specified in 3.4.6.

4.6.8 Abrasion resistance. The haze and luminous transmittance of the coated specimens shall be determined before and after the abrasion test as specified in with Method 3022 of FED-STD-406. The abrasion test shall be performed as specified in Method 1092 of FED-STD-406 using CS10F calibrase wheels for one hundred cycles under a 500 gram load.

4.6.9 Adhesion test method. A tape 3M670 (3M Company or equal) is firmly applied to the test specimen. The tape is removed using a snap motion applied 90 degrees to the test specimen. Test specimens shall be subjected to humidity tests as specified in Method 507 of MIL-STD-810, for 120 hours (5 cycles) and then subjected to the adhesion test.

4.6.10 Prismatic deviation (all classes of visors).

4.6.10.1 Vertical and horizontal prismatic deviation. The prismatic deviation, both vertically and horizontally, shall apply to readings taken on the visor "as worn" (1.70 inches from the concave surface) when measured at point "C". The test equipment shall be a calibrated transit or telescope, a lens holder, and an appropriate illuminated target at 35-foot distance-or equivalent equipment. The lens shall meet the requirements specified, (see 3.4.9).

4.6.11 Spherical and cylindrical power (all classes of visors). The spherical and cylindrical power shall be determined with the same apparatus used for measuring prismatic deviation, if the test equipment is appropriate, or shall be determined with a special lensometer or vertometer which measures from -0.25 to +0.25 diopters with 0.01 divisions (instead of the usual lensometer or vertometer which measures from -20.0 to +20.0 diopters with 0.12 divisions). The visor shall meet the requirements specified. (See 3.4.10).

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4.6.12 Critical vision area distortion (all classes of visors). With the visor inserted into the apparatus described in Figure 3 and its surface normal to the line of sight, examine for distortion. The degree of off-parallelism will consitute the amount of distortion and this distortion pattern shall be compared with Figure 2. The visor shall meet the requirements specified. (See 3.4.11).

4.6.13 Crack propagation resistance (thermal relocation method).

4.6.13.1 Specimen. The test specimen shall be taken from a completely formed visor and shall be 3.00 ±0.05 inches square by the thickness of the visor. Specimens may be taken from visors otherwise rejected for physical or optical defects as long as the processing to which they have been subjected is identical to that of production parts.

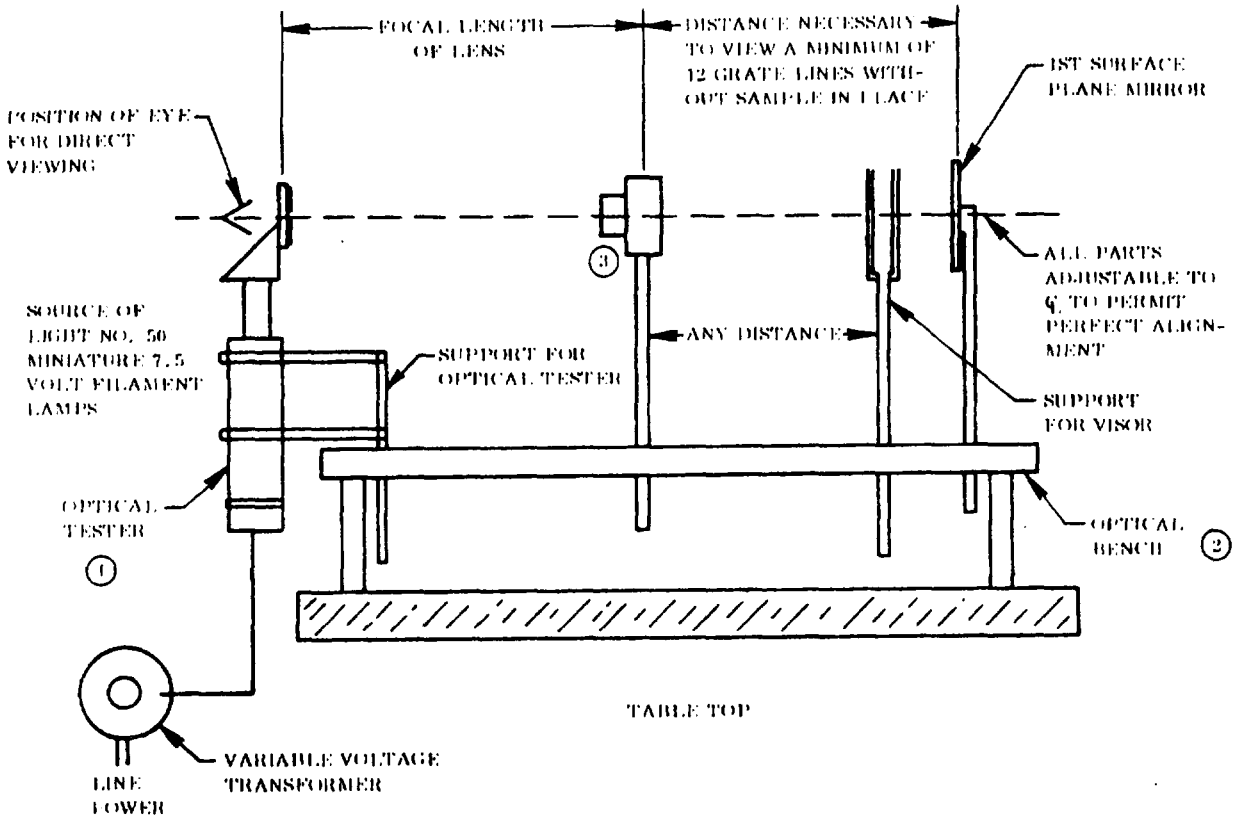
4.6.13.2 Method. Each specimen shall be scribed with a circle 2.00 ±0.1 inches in diameter and two perpendicular diameter lines therein. The scribed lines shall be oriented in the direction of maximum stretch and normal thereto. Diameters shall be measured to the nearest 0.001 inch at 73.5° ±2.0°F (23.0 ±1.1°C). Specimens shall be heated for 18 hours at 275° ±5°F (135° ±2.7°C) in a circulating air oven. At the end of this period the specimen shall be cooled to the temperature of the original measurements and again measured. Percent recovery shall be calculated for each diameter as follows and the two values averaged to obtain the recoverable stretch for the assembly. The visor shall meet the requirements specified. (See 3.4.12).

$$\text{Percent recoverable stretch} = \frac{\text{Initial length - Final length x 100}}{\text{Initial length}}$$

4.6.14 Impact resistance. A 5/8-inch steel ball weighing approximately 0.56 ounce shall be dropped from a height of 50 inches upon the upper horizontal surface of the visor. The steel ball shall strike within the 5/8 inch diameter (geometric center) of the visor. The visor itself shall be supported by a rigid acrylic plastic or steel tube 1 ± 1/8 inch on the inside diameter, 1-1/4 ±1/8 inches on the outside diameter and approximately 1-inch in height. The top edge of the tube shall have a 1/8 inch neoprene gasket, having a hardness of 40 ±5 on the durometer scale and a minimum tensile strength of 1200 pounds as determined by ASTM Method D 412, with a minimum elongation of 400 percent as determined by ASTM Method D 412. The diameter of the visor support may be modified so that the 1/8 inch neoprene gasket may support the visor at its periphery. Tha ball may be guided in its fall by being dropped through a tube, extending to within 4 inches of the visor. The visor shall meet the requirements specified. (See 3.4.13).

4.6.15 Penetration resistance. The visor shall be supported on a wooden block of such size and shape as to fit the visor securely. A pointed projectile of suitable size, consisting of a new Singer number 25, size 135 x 17 needle, fastened into a holder weighing approximately 1.56 ounces, shall be freely dropped, pointed downward, from a height of 50 inches onto the

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1. MODEL "B" OPTICAL TESTER WITH A 60-LINE GRATING, WITH MODEL "O" OPTICAL BENCH ADAPTER OR EQUIVALENT. (OBTAINABLE FROM ANN ARBOR OPTICAL CO, P.O. BOX 2056, ANN ARBOR, MICHIGAN)

2. OPTICAL BENCH OBTAINABLE FROM CENTRAL SCIENTIFIC CO, OR ITS EQUIVALENT OBTAINABLE FROM ANY EQUIPMENT SUPPLIER

3. TELEPHOTO CAMERA LENS—A SCHNEIDER TELE-ARTON 1:5.5/240 mm, NORMAL COLOR COATED, BARREL ONLY OR EQUAL

FIGURE 3. New London-Ann Arbor distortion tester No. 121959.

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horizontal outer surface of the visor. The projectile may be guided, but not restricted in its fall by being dropped through a tube extending to within approximately 4 inches of the visor. The visor shall not be pierced through from the impact.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified by the procuring activity (see 6.2).

5.1.1 Level A. Each visor shall be wrapped in flannel and packaged snugly in a unit container conforming to PPP-B-676.

5.1.2 Level C. Each visor shall be preserved and packaged to afford the minimum degree of protection necessary to prevent deterioration or damage during shipment under normal environmental conditions and commercial modes of transportation.

5.2 Packing. Packing shall be Level A, B, or C, as specified by the procuring activity (see 6.2). Shipping containers, shall be uniform in size and shape and of minimum cube and tare weight.

5.2.1 Level A. The visor packaged in the cartons as specified in 5.1.1 shall be packed in fiberboard containers conforming to PPP-B-636, Type II, Class 2, Grade 3. Closure and strapping of the containers shall be as specified in the container specification.

5.2.2 Level B. Visors packaged as specified in 5.1.1, shall be packed in fiberboard containers conforming to PPP-B-636, Type II, Class 1, Grade 3. Closure and strapping of the containers shall be as specified in the container specification. The gross weight of the packed shipping container shall not exceed the weight limitation of the container specification.

5.2.3 Level C. The visors which require packing for acceptance by the carrier, shall be packed within exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. The shipment shall conform to the minimum requirements of the rules and regulations applicable to the mode of transportation selected.

5.3 Marking. Interior and exterior containers shall be marked as specified in MIL-STD-129.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The visors covered by this specification are intended for use with aircrew protective helmet visor assemblies.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

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a. Title, number, and date of this specification
b. Class of visor and applicable drawing (see 1.2 and 3.4).
c. Number of first article samples to be submitted if other than specified 4.3.1.
d. Visors subjected to destructive tests (see Table I) shall not be shipped or considered as part of the contract or order.
e. Selection of applicable levels of preservation, packaging and packing.

6.2.2 Data requirements. For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order.

Preparing Activity

Navy (AS)

(Project No. 8475-N165)

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NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-V-85374(AS) VISORS, SHATTER RESISTANT

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

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EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

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